Module 7

Capital Budgeting

FINE 3010, Financial Management

Prof. Renping Li, Tulane University

Goals

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- How to decide what project(s) to do? The tool is the NPV rule
- But to use NPV, we need a project's cash flows
- Here, combine what we know about accounting and cash flow effects (**Module 2**) with the NPV rule (**Module 6**), to decide whether or not to accept a project
- **Capital budgeting**, the process of analyzing capital investment opportunities ("projects") and deciding which ones to accept

Roadmap

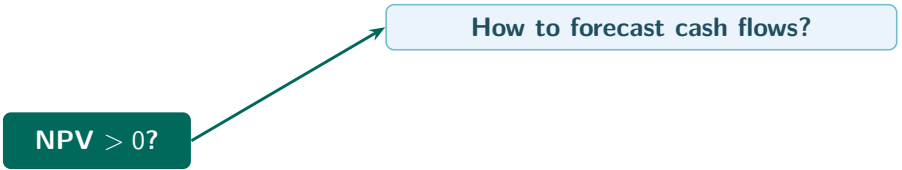
Should we invest in a project (e.g., a data center)?

NPV > 0?

Recall from **Module 6**

Roadmap

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How to forecast cash flows?

- Similar to **Module 2**, indirect method for cash flow
- But different, cash flow from a project, not a whole firm

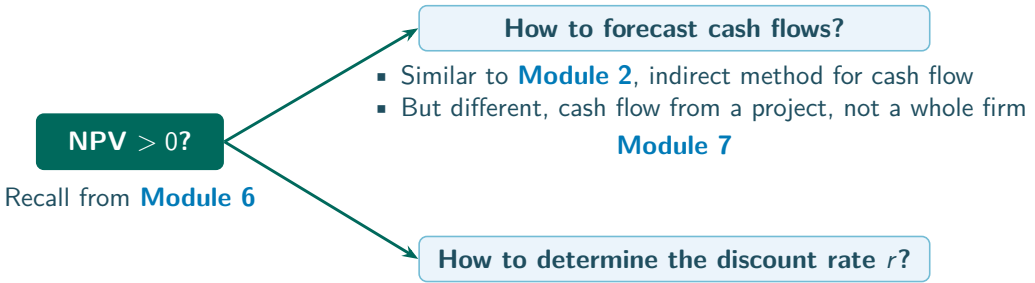
Module 7

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Module 7

How to determine the discount rate r?

Opportunity cost, the return from an investment with a similar level of risk

- What is risk?
- How is return related to risk?

Module 8

Motivating Example, Big Tech's AI Spending



Reuters, October 19, 2026 (5:54), "Big tech risks outspending its AI returns"

Module 7, Part 0

Review of the Statement of Cash Flows

Statement of Cash Flows

Recall from **Module 2**

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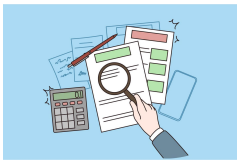
Recall from **Module 2**

Direct method	Indirect method
Lists every cash receipt and payment	Starts from net income and adjusts it
More time-consuming to prepare	Faster to prepare
Rarely used in practice	**The common choice**

Direct →



Indirect →



Statement of Cash Flows

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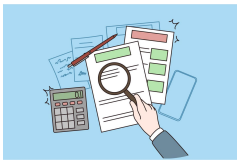
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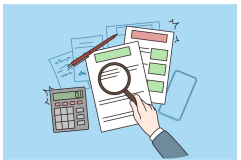
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- Both methods give the same totals
- We will follow the indirect method, starting from net income in the income statement

Direct →



Indirect →



How Do Net Income and Cash Flow Differ?

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Cash flow ($ billions)	**−2.0**	0	0	0	0	0	
Effect on net income	0	**−0.4**	**−0.4**	**−0.4**	**−0.4**	**−0.4**	← Depreciation

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When spending on fixed assets, CF = NI − spending on fixed assets

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When spending on fixed assets, CF = NI − spending on fixed assets
When recording depreciation, CF = NI + depreciation

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When raising money, CF = NI + amount raised

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Recall from **Module 2**

- The firm spends $60 million producing AI control panels
- None of them have been sold yet



OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

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Cash flow (\$ millions)	**−60**
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Recall from **Module 2**

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When firm pays off suppliers, CF = NI + change in accounts payable

Operating Activities

Recall from **Module 2**

Operating activities, cash from running the business day to day

Steps to calculate cash flow from operating activities,

1. **Start** with net income
2. **Add** depreciation
3. **Subtract** change in net working capital (NWC)
   - NWC = inventories + accounts receivable − accounts payable
   - NWC, cash tied up in day-to-day operations, in unsold goods and unpaid bills

Why Subtract Change in Net Working Capital (NWC)?

Recall from **Module 2**

When adding to inventories, CF = NI − change in inventories

When firm not yet paid by customers, CF = NI − change in accounts receivable

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$$CF = NI - \text{change in (inventories} + \text{accounts receivable} - \text{accounts payable)}$$

$$CF = NI - \text{change in net working capital (NWC)}$$

Operating Activities

Recall from **Module 2**

Cash flow from operating activities	
Net income	3,160
Depreciation	3,000
Change in net working capital	−1,000
Cash flow from operating activities	7,160

$$3{,}160 + 3{,}000 - (-1{,}000) = 7{,}160$$

Investing Activities

Recall from **Module 2**

Investing activities, cash spent buying fixed assets and cash received from selling them

Investing activities show cash flows from,

1. Capital purchased and sold
   - **Capital**, a firm's fixed assets
   - **Capital expenditure**, the amount spent on buying capital during the period
   - Cash flow is *negative* if firms buy capital

Investing Activities

Recall from **Module 2**

Cash flow from investing activities	
Cash flow from capital expenditure	−3,000
Cash flow from sales of fixed assets	200
Cash flow from investing activities	−2,800

$$-3{,}000 + 200 = -2{,}800$$

Financing Activities

Recall from **Module 2**

Financing activities, cash raised from lenders and shareholders, and cash returned to them

Recap of Part 0

Recall from **Module 2**

- The statement of cash flows, cash receipts and payments over a period
- Net income $\neq$ cash flow
- The statement has three parts
  - **Operating activities**, cash flow = Net income + Depreciation − Change in NWC
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them
  - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

Module 7, Part 1

Project Cash Flow

Firm Cash Flow vs. Project Cash Flow

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Firm Cash Flow vs. Project Cash Flow

- **Module 2** looked at the whole firm, here we look at one project on its own
- **Operating activities**, the same idea, project net income plus depreciation minus the change in NWC
- **Investing activities**, the project's capital investments, buying its assets at the start and selling them at the end
- **Financing activities**, *none*, we do not consider financing activities when evaluating a project on its own

Identify Project Cash Flow

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Cash flow from a project	= Project net income	Component 1
	+ Depreciation	Component 2
	+ Cash flow from changes in working capital	Component 3
	+ Cash flow from capital investments	Component 4

Component 1, Project Net Income

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Revenue from the project
− Cash expenses for the project
− Depreciation for the project's capital investment
Project EBIT
− Taxes (tax rate $\times$ project EBIT)
Project net income

Component 2, Depreciation

- Depreciation is subtracted on the project's income statement, but it is not a cash flow, the cash went out when the project's capital investment was made

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- Depreciation is subtracted on the project's income statement, but it is not a cash flow, the cash went out when the project's capital investment was made
- To get cash flow from the project, add it back, as in **Module 2**

Example, Project Net Income and Depreciation

- A project requires an investment of $2,000. The project's revenues minus its cash expenses are $1,100 in year 1 and $1,300 in year 2. The investment is depreciated at $1,000 per year. The corporate tax rate is 21%

Example, Project Net Income and Depreciation

- A project requires an investment of $2,000. The project's revenues minus its cash expenses are $1,100 in year 1 and $1,300 in year 2. The investment is depreciated at $1,000 per year. The corporate tax rate is 21%
- Calculate the project's net income plus depreciation for year 1 (Components 1 and 2)

Example, Project Net Income and Depreciation

- Solution,

Item	$
Revenues − Cash expenses	1,100
− Depreciation	1,000
Project EBIT	100
− Taxes	21
Project net income	79
+ Depreciation	1,000
Project net income + Depreciation	1,079

Example, Project Net Income and Depreciation

- Solution,

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- Taxes, $21\% \times$ project EBIT $= 21\% \times 100 = 21$

Example, Project Net Income and Depreciation

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Project net income + Depreciation	1,079

- Taxes, $21\% \times$ project EBIT $= 21\% \times 100 = 21$
- Project net income + Depreciation $= 79 + 1{,}000 = \$1{,}079$, Components 1 and 2 of the year-1 cash flow

Component 3, Cash Flow from Changes in Working Capital

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Component 3, Cash Flow from Changes in Working Capital

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- A project has its own NWC, e.g., an inventory of raw materials before production starts, and accounts receivable rising as sales start
- As in Module 2, an increase in the project's NWC is a negative cash flow, and a decrease is a positive cash flow,

  Cash flow from changes in working capital = − Change in the project's NWC

Component 3, Cash Flow from Changes in Working Capital

- E.g., a 3-year project needs $100 of NWC (raw materials in stock, customers' bills not yet paid) while it operates, recovered when the project ends

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Year	0	1	2	3
The project's NWC	100	100	100	0
Change in NWC	$+100$	0	0	-100
Cash flow from changes in working capital	-100	0	0	$+100$

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Year	0	1	2	3
The project's NWC	100	100	100	0
Change in NWC	+100	0	0	−100
Cash flow from changes in working capital	−100	0	0	+100

- Cash goes out up front, as the project stocks up and waits to be paid, and comes back at the end, when the stock is sold and the bills are collected

Component 3, Cash Flow from Changes in Working Capital

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Year	0	1	2	3
The project's NWC	100	100	100	0
Change in NWC	+100	0	0	−100
Cash flow from changes in working capital	−100	0	0	+100

- Cash goes out up front, as the project stocks up and waits to be paid, and comes back at the end, when the stock is sold and the bills are collected
- Undiscounted the two cancel, but the \$100 goes out at $t = 0$ and returns only at $t = 3$, so tying up the cash still costs the time value of money

Component 4, Cash Flow from Capital Investments

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- Selling the project's assets at the end is a positive cash flow
- The investment is usually depreciated over time, in this course always "straight-line" depreciation,

$$\text{Annual depreciation} = \frac{\text{Investment} - \text{Salvage value}}{\text{Expected life}}$$

Recap of Part 1

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- The project's capital investments are negative cash flows when the assets are bought and positive when they are sold, depreciated straight-line over the project's life

Module 7, Part 2

Putting It All Together

Example, Putting It All Together

- A company is considering building a new data center which costs \$1 billion (\$1,000 million) and will last 5 years. It will be depreciated straight-line over 5 years with a salvage value of \$200 million. The data center can also be sold at $t = 5$ for \$200 million

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- Should the company build the data center or not?

Example, Putting It All Together

- Solution ($ millions),

Year	0	1	2	3	4	5
Revenues − Cash expenses		200	200	200	200	200
Depreciation		160	160	160	160	160
EBIT		40	40	40	40	40
Net income		31.6	31.6	31.6	31.6	31.6
+ Depreciation		160	160	160	160	160
Operating cash flows		191.6	191.6	191.6	191.6	191.6
Capital investment	−1,000					
Sale of the data center						200
Working capital	−100					100
Total cash flows	−1,100	191.6	191.6	191.6	191.6	491.6
NPV	−130.82					

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  - **Operating cash flow before accounting for ΔNWC**, in years 1–5 the rental income adds 200 to revenues minus cash expenses. Depreciation rises by $(1{,}000 - 200)/5 = 160$ per year, a net 40 higher EBIT. After taxes and adding back depreciation, operating cash flow rises by 191.6 each year

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- Putting it all together, NPV = −$130.82 million, so the company should not build the data center

Module 7, Part 3

The Principles behind Project Cash Flows

General Principles for Identifying Project Cash Flow

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- **Accounting approach** (the **wrong** approach, it can lead to erroneous decisions), consider accounting income
- In the data center table, we discounted the project's operating cash flows ($191.6 million a year), not its net income ($31.6 million)

Cash Flow vs. Accounting Income

- A project costs $2,000 and is expected to last 2 years, producing cash income of $1,500 and $500, respectively. The cost can be depreciated at $1,000 per year. Given a 10% required return, compare the NPV using accounting income to the NPV using cash flow

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- This project is clearly a loser!

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- The data center table has no interest line, even if the company borrows to build it

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- Different depreciation schedules, in this course we stick to "straight-line" depreciation, as in the data center, $(1{,}000 - 200)/5 = \$160$ million a year

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$$\text{Incremental cash flow} = \text{Cash flow with the project} - \text{Cash flow without it}$$

- In the data center table, the rental income and the spare parts exist only because of the data center, so they count

Incremental Cash Flows, a Few Rules That Are Easy to Forget

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f. Remember terminal cash flows

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- What is the **incremental** cash flow of the new console?





Rule a, Indirect Effects

- Solution ($ millions),

	$t = 0$	$t = 1$	$t = 2$
Cash flow from the old console without the new one		800	1,000
Cash flow from the old console with the new one		500	400
Cash flow from investment in and sales of the new one	-200	400	800
Incremental cash flow of the new console	-200	$400 - 300 = 100$	$800 - 600 = 200$

Rule b, Sunk Costs

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- Many sunk costs are "related" to a project, but no longer incremental if the current decision does not affect them



Example, Sunk Costs

- Your company has developed a new product that is almost ready to launch. The R&D cost $0.6 million and was completed last year
- Launching requires another $1 million investment (marketing and so on), with a present value of future operating cash flows of $1.5 million

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- Launching requires another $1 million investment (marketing and so on), with a present value of future operating cash flows of $1.5 million
- Is this project a good idea today?
- Was this project a good idea last year?

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- What is the actual cost of building the factory?

Rule d, Overhead

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- The principle of incremental cash flows says only *changes* in overhead that result from a project count

Rule e, Working Capital

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  - Easy to forget! (The data center example had them)
  - Only changes in NWC that are incremental to the project count
- Working capital often keeps changing during the project, and is often recovered at the end
- Common mistakes,
  1. Forgetting about working capital entirely
  2. Forgetting that working capital may change during the life of the project
  3. Forgetting that working capital is often recovered at the end of the project

Rule f, Terminal Cash Flows

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$$(500 - 200) \times 21\% = \$63 \text{ million}$$

- The net (after-tax) proceeds are $500 - 63 = \$437$ million, and now the NPV is positive ($+\$30.48$ million)

Example, the Data Center with a Higher Terminal Cash Flow

- Solution ($ millions),

Year	0	1	2	3	4	5
Revenues − Cash expenses		200	200	200	200	200
Depreciation		160	160	160	160	160
EBIT		40	40	40	40	40
Net income		31.6	31.6	31.6	31.6	31.6
+ Depreciation		160	160	160	160	160
Operating cash flows		191.6	191.6	191.6	191.6	191.6
Capital investment	−1,000					
Sale of the data center (after tax)						437
Working capital	−100					100
Total cash flows	−1,100	191.6	191.6	191.6	191.6	728.6
NPV	30.48					
IRR	8.8%					

Rule f, Terminal Cash Flows Can Be Negative

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- E.g., oil wells require "capping" when they no longer produce. Capping is expensive, so oil well projects are worth more if you do not cap them (although this is illegal!)



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- E.g., oil wells require "capping" when they no longer produce. Capping is expensive, so oil well projects are worth more if you do not cap them (although this is illegal!)
- But apparently many companies still do this, Bloomberg, "It Pays to Not Pay Your Debts"



Practice Problem

- A firm plans a new manufacturing plant on a site it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. The market value of the site
b. The market value of the existing buildings
c. Demolition costs and site clearance
d. The cost of a new access road put in last year
e. Lost cash flows on other projects due to executive time spent on the new facility
f. Future depreciation of the new plant
g. The reduction in the firm's tax bill resulting from depreciation of the new plant
h. The initial investment in inventories of raw materials
i. Money already spent on engineering design of the new plant

Practice Problem

a, b. The site and buildings could have been sold or put to another use. Their values are opportunity costs, treated as incremental cash outflows

Practice Problem

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- Discount cash flows, not accounting profits, and ignore financing (all-equity)

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- Count indirect effects, opportunity costs, working capital, and terminal cash flows (with the tax on any gain over book value), ignore sunk costs and allocated overhead

Module 7, Part 4

Full-Scale Capital Budgeting Problems

Example, the Data Center Again

- Recall the data center from Part 2 ($ millions),
  - It costs 1,000 and lasts 5 years, depreciated straight-line to a salvage value of 200, and sold at $t = 5$ for 200

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- This time, build the table step by step

Example, the Data Center Again

- Solution ($ millions), the operating cash flows first, then working capital, then the capital investment and the sale, then the totals

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Revenues − Cash expenses		200	200	200	200	200
Depreciation		160	160	160	160	160
EBIT		40	40	40	40	40
Net income		31.6	31.6	31.6	31.6	31.6
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Working capital	−100					100
Capital investment	−1,000					
Sale of the data center						200
Total cash flows	−1,100	191.6	191.6	191.6	191.6	491.6
NPV	−130.82					

- NPV = −$130.82 million, reject the project, the same answer as in Part 2

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4. Total the cash flows year by year, discount at the discount rate, and apply the NPV rule

Example 1, Growing Sales and a Taxed Sale of Equipment

- A company makes bottles and just developed a new bottle design
- The initial investment in machinery is \$6 million. The equipment is depreciated straight-line over 6 years to zero, but will actually be sold in 6 years for \$620,000 (so there are tax consequences)

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- The new design is phased out after 6 years. The tax rate is 21%, and the required rate of return is 12%
- What is the project's NPV?

Example 1, Growing Sales and a Taxed Sale of Equipment

- Operating cash flows ($ thousands),

Year	1	2	3	4	5	6
Sales (thousand bottles)	1,000	1,050	1,103	1,158	1,216	1,276
Sales ($)	4,000	4,200	4,410	4,631	4,862	5,105
Cost ($)	2,000	2,100	2,205	2,315	2,431	2,553
Depreciation	1,000	1,000	1,000	1,000	1,000	1,000
EBIT	1,000	1,100	1,205	1,315	1,431	1,553
Taxes	210	231	253	276	301	326
Net income	790	869	952	1,039	1,130	1,227
+ Depreciation	1,000	1,000	1,000	1,000	1,000	1,000
Operating cash flows	1,790	1,869	1,952	2,039	2,130	2,227

Example 1, Growing Sales and a Taxed Sale of Equipment

- Terminal cash flow from the sale of the equipment ($ thousands), the book value is zero so the whole sale price is taxed,

Terminal cash flows	$
Sales price	620
Tax	130
Net income from sale	490

Example 1, Growing Sales and a Taxed Sale of Equipment

- Total cash flows ($ thousands),

Year	0	1	2	3	4	5	6
Capital	−6,000						490
Working capital	−250						250
Operating cash flows		1,790	1,869	1,952	2,039	2,130	2,227
Total cash flows	−6,250	1,790	1,869	1,952	2,039	2,130	2,966
PV	−6,250	1,598	1,490	1,389	1,296	1,209	1,503
NPV	2,235						

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PV	−6,250	1,598	1,490	1,389	1,296	1,209	1,503
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- The Excel solution is in the "Bottles" tab of Module 7 Excel Example.xlsx

Practice Problem

- A firm has developed a new mousetrap. It can go into production for an initial investment in equipment of \$6.3 million. The equipment will be depreciated straight-line over 6 years, but in fact can be sold after 6 years for \$695,000
- Working capital at each date must be maintained at 10% of next year's forecast sales. Production costs are \$1.80 per trap and the traps sell for \$7 each. The project ends in 6 years when the trap becomes obsolete. The tax rate is 40%, and the required rate of return is 11%

Year	0	1	2	3	4	5	6	After
Sales (million traps)	0	0.6	0.8	0.9	0.9	0.5	0.2	0

Practice Problem

- Suppose the firm can cut its working capital requirements in half with better inventory control. By how much does this increase the project NPV?

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- Solution ($ millions), working capital at each date is 10% of next year's sales,

Year	0	1	2	3	4	5	6
Sales units		0.600	0.800	0.900	0.900	0.500	0.200
Revenue	0.000	4.200	5.600	6.300	6.300	3.500	1.400
NWC	0.420	0.560	0.630	0.630	0.350	0.140	0.000
Cash flow from NWC	−0.420	−0.140	−0.070	0.000	0.280	0.210	0.140

Practice Problem

- With working capital requirements halved ($ millions),

Year	0	1	2	3	4	5	6
New NWC	0.210	0.280	0.315	0.315	0.175	0.070	0.000
New cash flow from NWC	−0.210	−0.070	−0.035	0.000	0.140	0.105	0.070
Change in cash flow	0.210	0.070	0.035	0.000	−0.140	−0.105	−0.070

Practice Problem

- With working capital requirements halved ($ millions),

Year	0	1	2	3	4	5	6
New NWC	0.210	0.280	0.315	0.315	0.175	0.070	0.000
New cash flow from NWC	−0.210	−0.070	−0.035	0.000	0.140	0.105	0.070
Change in cash flow	0.210	0.070	0.035	0.000	−0.140	−0.105	−0.070

- Discounting the changes in cash flow at 11%, the change in NPV is $0.1095 million

Practice Problem

- A firm may upgrade its modem pool. It last upgraded 2 years ago, when it spent $140 million on equipment with an assumed life of 5 years and an assumed salvage value of $20 million for tax purposes. The firm uses straight-line depreciation. The old equipment can be sold today for $100 million
- A new modem pool can be set up today for $210 million. It has a 3-year life and is depreciated to zero straight-line. The new equipment increases sales by $28 million per year and decreases operating costs by $14 million per year. At the end of 3 years it is worthless. The tax rate is 30% and the discount rate is 9%

Practice Problem

a. What is the net cash flow at time 0 if the old equipment is replaced?
b. What are the incremental cash flows in years 1, 2, and 3?
c. What is the NPV of the replacement project?
d. What is the IRR of the replacement project?

Practice Problem

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Sales price = $100, so net-of-tax proceeds from the sale are:
$100 − [0.30 × ($100 − $92)] = $97.60

Therefore, the net cash outlay at time 0 is $210 − $97.60 = −$112.40.

b. The project saves $14 in operating costs and increases sales by $28. The depreciation expense for the new machine would be $70 per year.

Therefore, including the depreciation tax shield, operating cash flow increases by:
($28 + $14) × (1 − 0.30) + ($70 × 0.30) = $50.40 per year

c. NPV = −$112.40 + $50.40 × ((1 ÷ 0.09) − {1 ÷ [0.09(1 + 0.09)^3]})
= $15.18

d. To find the internal rate of return, find the interest rate "r" that makes the Present Value of the inflows equal and opposite to the initial outflow. To do this set the PV of the three year annuity to $112.40 and solve for the discount rate (r):

$112.40 = $50.40 × [1 ÷ r − 1 ÷ r × (1 + r)^3]
r = 16.43%

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- Lay the years out in a table, total the cash flows, discount, and apply the NPV rule

Thank you!

Please let me know if you have any questions.